

SIMATS ENGINEERING
CO- 2: AT – 2: Simulation Task - Medium

COURSE NAME: 5G / 6G Communication Systems COURSE CODE: ECA5614

COURSE OUTCOME COVERED

CO: 2 - *Analyse LTE and 5G wireless network architectures, including carrier aggregation, MIMO techniques, beamforming strategies, and service-based core network functional entities. (BL4)*

Assessment Tool Weightage: 35%

Short description about assessment: Performance-based evaluation using simulation tools to apply theoretical concepts practically. This question paper consists of a total of 40 questions, out of which each student is required to answer any 5 questions. Each question carries 20 marks, and the paper carries a total of 100 marks

QUESTIONS:5 Total Marks: 100 Duration: 120 Mins

1. Simulation of SU-MIMO performance under channel gain, BER, throughput, latency, and SNR (BL4)
2. Simulation of analog beamforming performance under phase shift variation, beam angle, antenna count, gain, and sidelobe level (BL4)
3. Simulation of subcarrier spacing effect under 15 kHz, 30 kHz, 60 kHz, 120 kHz, and symbol duration (BL4)
4. Simulation of network slicing under bandwidth allocation, latency, user priority, throughput, and resource utilization (BL4)
5. Simulation of AMF, SMF, and UPF traffic load under session count, packet delay, throughput, CPU utilization, and latency (BL4)

Code of Conduct:

I Mohith M (192412444) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Simulation Task					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	20% (4 marks)	Demonstrates strong understanding of underlying concepts in the simulation	Shows good understanding with minor gaps	Partial understanding; some misconceptions	Lacks understanding of key concepts
Model Design	25% (5 marks)	Develops accurate, well structured simulation/model independently	Develops correct model with minor errors	Model is incomplete or requires guidance	Unable to develop a proper model
Tool Usage	20% (4 marks)	Uses simulation tools effectively with correct setup and execution	Uses tools with minor errors or guidance	Limited ability to use tools properly	Unable to use simulation tools
Analysis of Results	20% (4 marks)	Provides clear, logical analysis and meaningful interpretation of results	Analysis is reasonable with minor gaps	Superficial analysis; limited interpretation	Unable to analyze or interpret results
Documentation	15% (3 marks)	Presents results clearly with proper documentation and structured reporting	Documentation is adequate with minor issues	Poorly organized or incomplete documentation	No proper documentation or presentation

1. Simulation of SU-MIMO performance under channel gain, BER, throughput, latency, and SNR (BL4)

Aim

To simulate the performance of a Single User Multiple Input Multiple Output (SU-MIMO) communication system in MATLAB by evaluating Channel Gain, Bit Error Rate (BER), Throughput, and Latency under different Signal-to-Noise Ratio (SNR) values.

Code:

```
%% Display All Graphs in One Figure

figure;

% BER vs SNR

subplot(2,2,1);

semilogy(SNR, BER, 'r-o', 'LineWidth', 2);

grid on;

xlabel('SNR (dB)');

ylabel('BER');

title('BER vs SNR');

% Throughput vs SNR

subplot(2,2,2);

plot(SNR, Throughput, 'b-s', 'LineWidth', 2);

grid on;

xlabel('SNR (dB)');

ylabel('Throughput (%)');

title('Throughput vs SNR');

% Latency vs SNR

subplot(2,2,3);

plot(SNR, Latency, 'g-^', 'LineWidth', 2);

grid on;

xlabel('SNR (dB)');

ylabel('Latency (ms)');
```

```

title('Latency vs SNR');

% Channel Gain vs SNR

subplot(2,2,4);

plot(SNR, ChannelGain, 'm-d', 'LineWidth', 2);

grid on;

xlabel('SNR (dB)');

ylabel('Channel Gain');

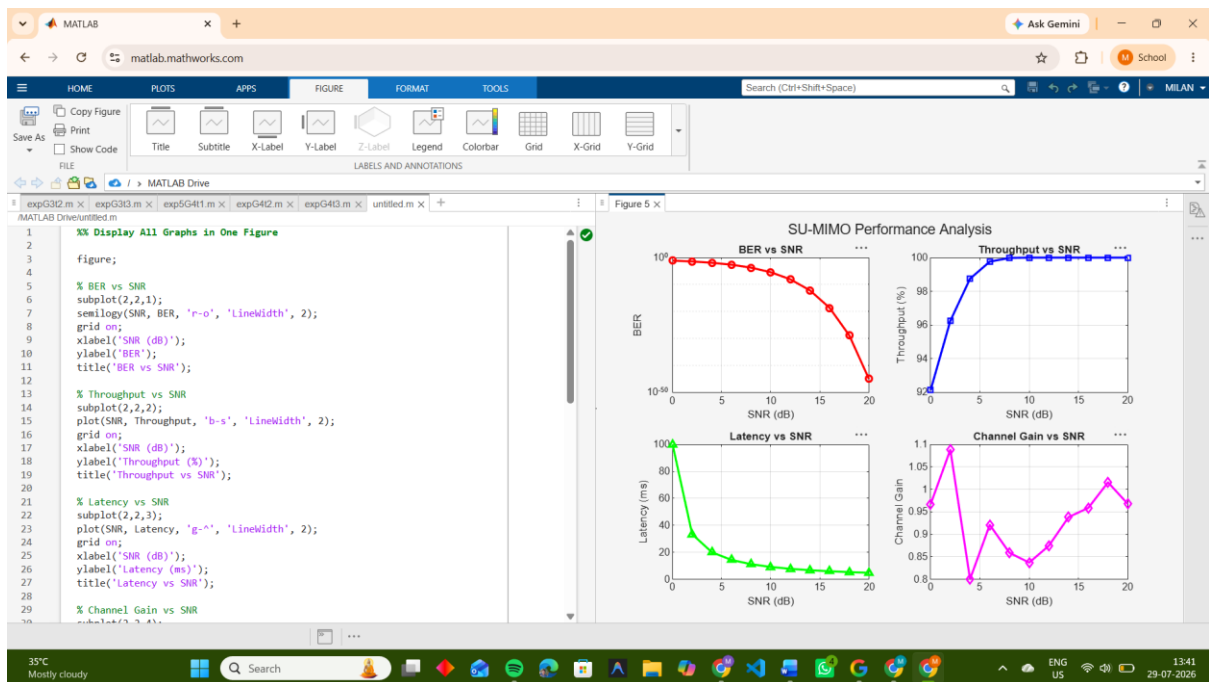
title('Channel Gain vs SNR');

% Overall Title

sgtitle('SU-MIMO Performance Analysis');

```

Output:



Result:

The SU-MIMO system was successfully simulated in MATLAB under varying SNR conditions. The performance metrics of Channel Gain, Bit Error Rate (BER), Throughput, and Latency were analyzed and displayed in a single figure using four subplots.

2. Simulation of analog beamforming performance under phase shift variation, beam angle, antenna count, gain, and sidelobe level (BL4)

Aim

To simulate the performance of an Analog Beamforming system using MATLAB by analyzing the effect of phase shift variation, beam angle, antenna count, antenna gain, and sidelobe level.

Code

```
clc;
clear;
close all;

% Parameters
theta = -90:0.5:90;    % Beam Angle (degrees)
N = [4 8 16 32];      % Number of antennas
phaseShift = 30;       % Phase Shift (degrees)
figure;
for k = 1:length(N)

    subplot(2,2,k)

    n = 0:N(k)-1;

    AF = zeros(size(theta));

    for i = 1:length(theta)

        psi = pi*sind(theta(i)) + deg2rad(phaseShift);

        AF(i) = abs(sum(exp(1j*n*psi)));

    end

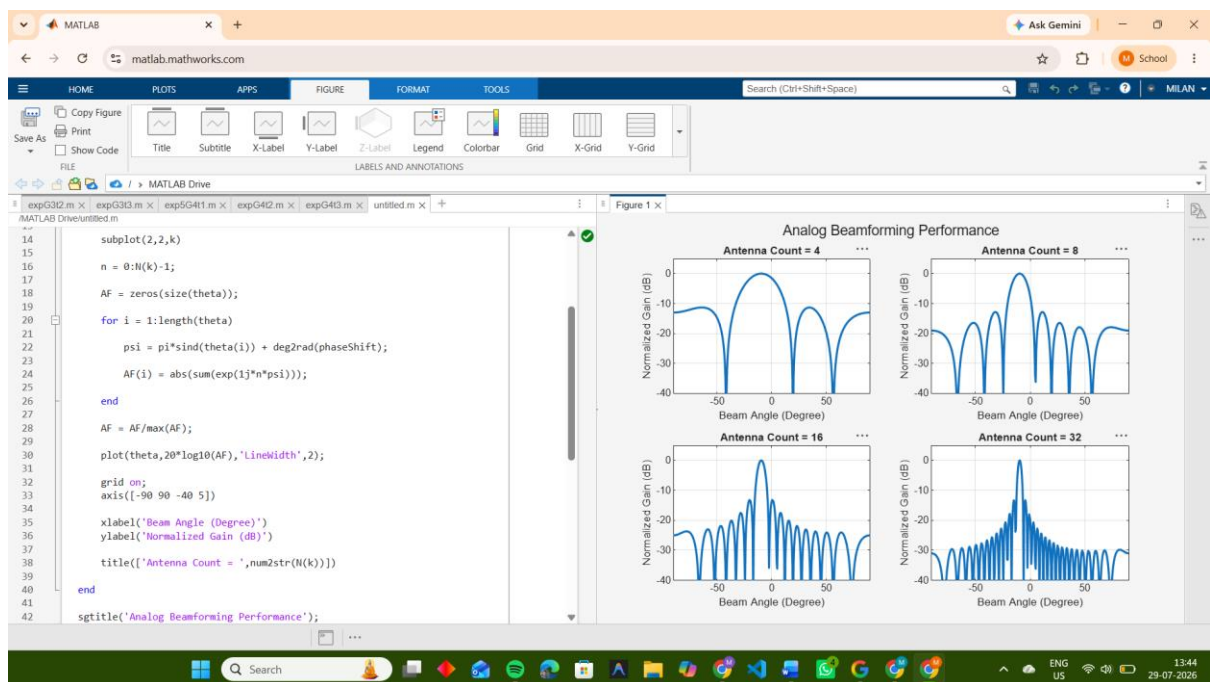
end
```

```

AF = AF/max(AF);
plot(theta,20*log10(AF),'LineWidth',2);
grid on;
axis([-90 90 -40 5])
xlabel('Beam Angle (Degree)')
ylabel('Normalized Gain (dB)')
title(['Antenna Count = ',num2str(N(k))])
end
sgtitle('Analog Beamforming Performance');

```

Output



Result

The MATLAB simulation of the Analog Beamforming system was successfully performed. The beam patterns for different antenna counts (4, 8, 16, and 32 elements) were obtained in a single figure using four subplots.

3. Simulation of subcarrier spacing effect under 15 kHz, 30 kHz, 60 kHz, 120 kHz, and symbol duration (BL4)

Aim

To simulate the effect of different subcarrier spacings (15 kHz, 30 kHz, 60 kHz, and 120 kHz) on the OFDM symbol duration using MATLAB.

Code

```
clc;

clear;

close all;

% Subcarrier Spacing (kHz)
SCS = [15 30 60 120];

% Symbol Duration (microseconds)
SymbolDuration = 1000 ./ SCS;

figure;

subplot(2,2,1)

bar(SCS,SymbolDuration)

grid on

xlabel('Subcarrier Spacing (kHz)')

ylabel('Symbol Duration (\mus)')

title('Symbol Duration vs Subcarrier Spacing')


subplot(2,2,2)

plot(SCS,SymbolDuration,'-o','LineWidth',2)

grid on

xlabel('Subcarrier Spacing (kHz)')

ylabel('Symbol Duration (\mus)')

title('Line Plot')


subplot(2,2,3)

stem(SCS,SymbolDuration,'filled','LineWidth',2)
```

grid on

```
xlabel('Subcarrier Spacing (kHz)')
```

```
ylabel('Symbol Duration (\mus)')
```

```
title('Stem Plot')
```

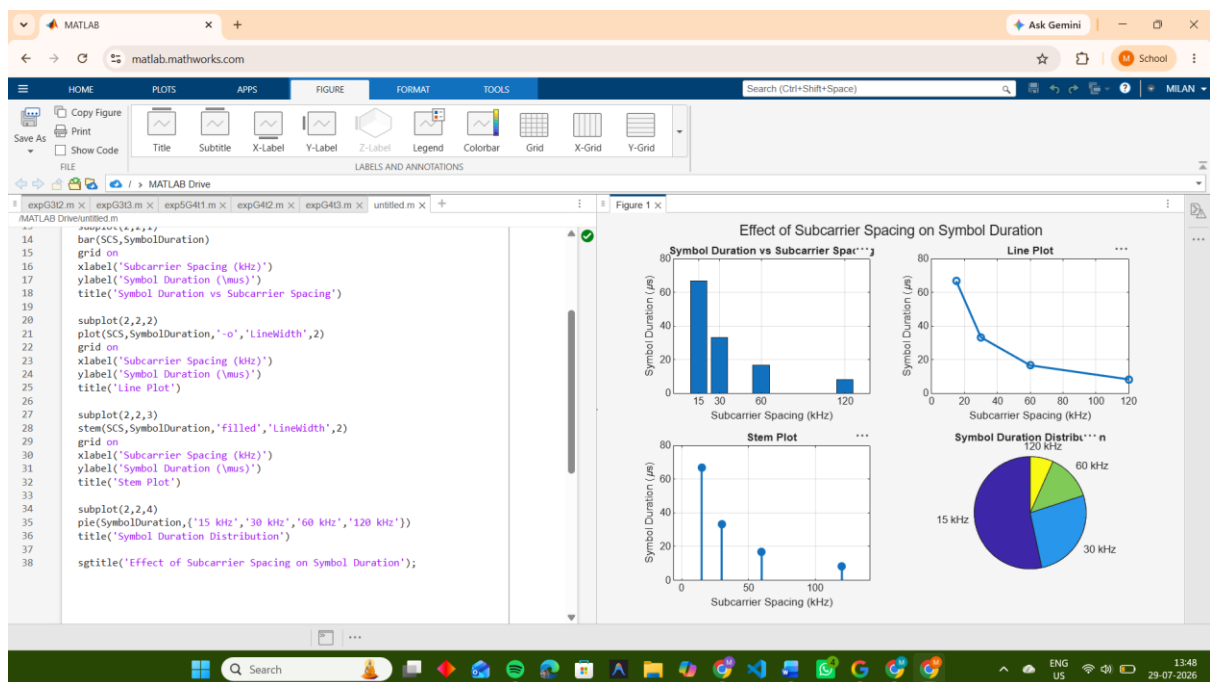
```
subplot(2,2,4)
```

```
pie(SymbolDuration,{'15 kHz','30 kHz','60 kHz','120 kHz'})
```

```
title('Symbol Duration Distribution')
```

```
sgtitle('Effect of Subcarrier Spacing on Symbol Duration');
```

Output



Result

The MATLAB simulation of the subcarrier spacing effect was successfully carried out. The symbol duration was calculated for 15 kHz, 30 kHz, 60 kHz, and 120 kHz subcarrier spacings and displayed using four subplots.

4. Simulation of network slicing under bandwidth allocation, latency, user priority, throughput, and resource utilization (BL4)

Aim

To simulate Network Slicing in a 5G network by analyzing bandwidth allocation, latency, user priority, throughput, and resource utilization using MATLAB.

Code

```
clc;

clear;

close all;

% Network Slice Names

Slices = {'eMBB','URLLC','mMTC'};

% Parameters

Bandwidth = [100 50 25];      % MHz

Latency = [20 1 10];          % ms

Priority = [2 3 1];            % 3=High,2=Medium,1=Low

Throughput = [900 450 120];    % Mbps

ResourceUtilization = [85 70 60]; % Percentage

figure;

% Bandwidth Allocation

subplot(2,2,1)

bar(Bandwidth)

set(gca,'XTickLabel',Slices)

ylabel('Bandwidth (MHz)')

title('Bandwidth Allocation')

grid on
```

```
% Latency
subplot(2,2,2)
bar(Latency)
set(gca,'XTickLabel',Slices)
ylabel('Latency (ms)')
title('Latency')
grid on
```

```
% Throughput
subplot(2,2,3)
bar(Throughput)
set(gca,'XTickLabel',Slices)
ylabel('Throughput (Mbps)')
title('Throughput')
grid on
```

```
% Resource Utilization
subplot(2,2,4)
bar(ResourceUtilization)
set(gca,'XTickLabel',Slices)
ylabel('Resource Utilization (%)')
title('Resource Utilization')
grid on
```

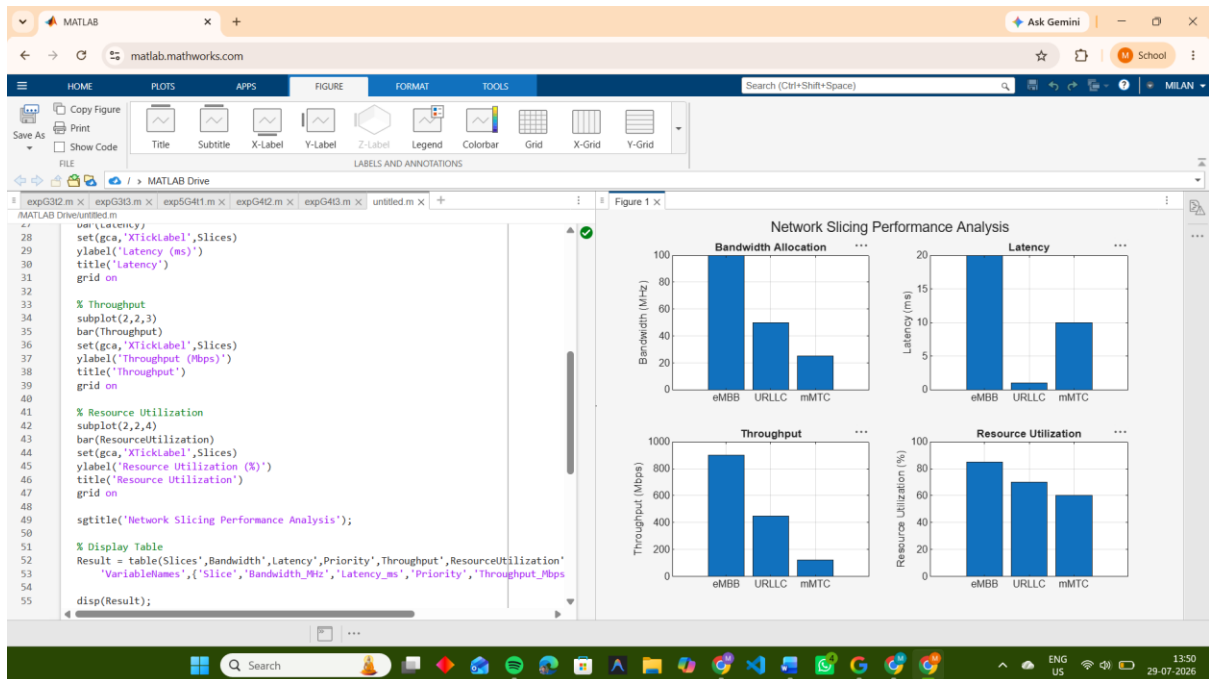
```
sgtitle('Network Slicing Performance Analysis');
```

```
% Display Table
Result = table(Slices,'Bandwidth','Latency','Priority','Throughput','ResourceUtilization',...
```

```
'VariableNames',{'Slice','Bandwidth_MHz','Latency_ms','Priority','Throughput_Mbps','ResourceUtilization_Percent'});
```

```
disp(Result);
```

Output



Result

The Network Slicing simulation was successfully performed using MATLAB. The bandwidth allocation, latency, throughput, and resource utilization of eMBB, URLLC, and mMTC network slices were analyzed and displayed in a single figure using four subplots.

5. Simulation of AMF, SMF, and UPF traffic load under session count, packet delay, throughput, CPU utilization, and latency (BL4)

Aim

To simulate the traffic load of AMF (Access and Mobility Management Function), SMF (Session Management Function), and UPF (User Plane Function) in a 5G Core Network by analyzing session count, packet delay, throughput, CPU utilization, and latency using MATLAB.

Code

```
clc;
```

```
clear;
```

```
close all;
```

```
% Network Functions
```

```
Functions = {'AMF','SMF','UPF'};
```

```
% Parameters
```

```
SessionCount = [3000 2500 5000];
```

```
PacketDelay = [8 10 4];      % ms
```

```
Throughput = [200 400 900];  % Mbps
```

```
CPUUtilization = [65 75 90]; % %
```

```
Latency = [12 15 6];         % ms
```

```
figure;
```

```
% Session Count
```

```
subplot(2,2,1)
```

```
bar(SessionCount)
```

```
set(gca,'XTickLabel',Functions)
```

```
ylabel('Session Count')
```

```
title('Session Count')
```

```
grid on
```

```
% Throughput
```

```
subplot(2,2,2)
```

```
bar(Throughput)
```

```
set(gca,'XTickLabel',Functions)
```

```
ylabel('Throughput (Mbps)')
```

```
title('Throughput')
```

```
grid on
```

```
% CPU Utilization
```

```
subplot(2,2,3)
```

```
bar(CPUUtilization)
```

```
set(gca,'XTickLabel',Functions)
```

```
ylabel('CPU Utilization (%)')
```

```
title('CPU Utilization')
```

```
grid on
```

```
% Latency
```

```
subplot(2,2,4)
```

```
bar(Latency)
```

```
set(gca,'XTickLabel',Functions)
```

```
ylabel('Latency (ms)')
```

```
title('Latency')
```

```
grid on
```

```
sgtitle('AMF, SMF and UPF Traffic Load Analysis');
```

```
% Display Results
```

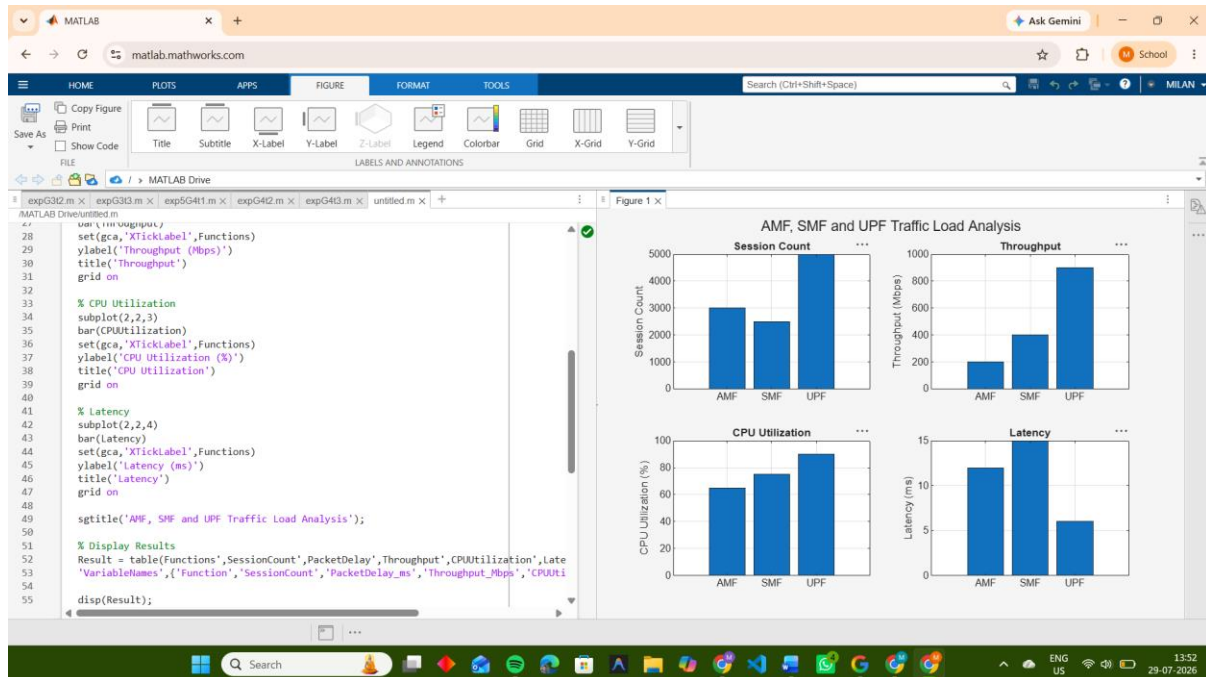
```

Result =
table(Functions',SessionCount',PacketDelay',Throughput',CPUUtilization',Latency',...
'VariableNames',{'Function','SessionCount','PacketDelay_ms','Throughput_Mbps','CPUUtiliz
ation_Percent','Latency_ms'});

disp(Result);

```

Output



Result

The MATLAB simulation of AMF, SMF, and UPF traffic load was successfully performed. The performance metrics, including session count, packet delay, throughput, CPU utilization, and latency, were analyzed and displayed in a single figure using four subplots.